

EXP14

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# 2. Load Dataset

housing = fetch_california_housing()
df = pd.DataFrame(
    housing.data,
    columns=housing.feature_names
)

df["Price"] = housing.target

print("Dataset:")
print(df.head())

# 3. Check Dataset

print("\nDataset Shape:", df.shape)
print("\nMissing Values:")
print(df.isnull().sum())

# 4. Separate Input and Output

X = df.drop("Price", axis=1)
y = df["Price"]

# 5. Train-Test Split

X_train, X_test, y_train, y_test = train_test_split(
```

```
X, y,  
test_size=0.20,  
random_state=42  
)
```

6. Create Model

```
model = LinearRegression()
```

7. Train Model

```
model.fit(X_train, y_train)
```

```
print("\nModel Training Completed!")
```

8. Prediction

```
y_pred = model.predict(X_test)
```

9. Model Evaluation

```
mae = mean_absolute_error(y_test, y_pred)
```

```
mse = mean_squared_error(y_test, y_pred)
```

```
rmse = np.sqrt(mse)
```

```
r2 = r2_score(y_test, y_pred)
```

```
print("\nMODEL EVALUATION")
```

```
print("-----")
```

```
print("MAE :", mae)
```

```
print("MSE :", mse)
```

```
print("RMSE :", rmse)
```

```
print("R2 Score :", r2)
```

10. Convert Prices to Indian Rupees

Dataset price is in units of \$100,000

```
# Approximate conversion: 1 USD = ₹85
```

```
actual_inr = y_test.values * 100000 * 85
```

```
predicted_inr = y_pred * 100000 * 85
```

```
# 11. Display Actual vs Predicted
```

```
comparison = pd.DataFrame({  
    "Actual Price (₹)": actual_inr[:10],  
    "Predicted Price (₹)": predicted_inr[:10]  
})
```

```
print("\nACTUAL VS PREDICTED PRICE")
```

```
print("-----")
```

```
print(comparison.round(2))
```

```
# 12. Plot Actual vs Predicted
```

```
plt.figure(figsize=(8, 5))
```

```
plt.scatter(actual_inr, predicted_inr)
```

```
plt.xlabel("Actual House Price (₹)")
```

```
plt.ylabel("Predicted House Price (₹)")
```

```
plt.title("Actual vs Predicted House Prices")
```

```
plt.show()
```

```
# 13. Predict Price for a New House
```

```
new_house = np.array([[
```

```
    5.0,    # MedInc
```

```
    20.0,   # HouseAge
```

```

5.0,    # AveRooms
1.0,    # AveBedrms
1000.0, # Population
3.0,    # AveOccup
34.0,   # Latitude
-118.0  # Longitude
]])

```

```
prediction = model.predict(new_house)
```

```
# Convert prediction to INR
```

```
price_usd = prediction[0] * 100000
```

```
price_inr = price_usd * 85
```

```
print("\nNEW HOUSE PRICE PREDICTION")
```

```
print("-----")
```

```
print("Predicted Price in USD : $", round(price_usd, 2))
```

```
print("Predicted Price in INR : ₹", round(price_inr, 2))
```

OUTPUT

Dataset:

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude \
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85

Longitude Price

0	-122.23	4.526
1	-122.22	3.585
2	-122.24	3.521

3 -122.25 3.413

4 -122.25 3.422

Dataset Shape: (20640, 9)

Missing Values:

MedInc 0

HouseAge 0

AveRooms 0

AveBedrms 0

Population 0

AveOccup 0

Latitude 0

Longitude 0

Price 0

dtype: int64

Model Training Completed!

MODEL EVALUATION

MAE : 0.5332001304956553

MSE : 0.5558915986952444

RMSE : 0.7455813830127764

R2 Score : 0.5757877060324508

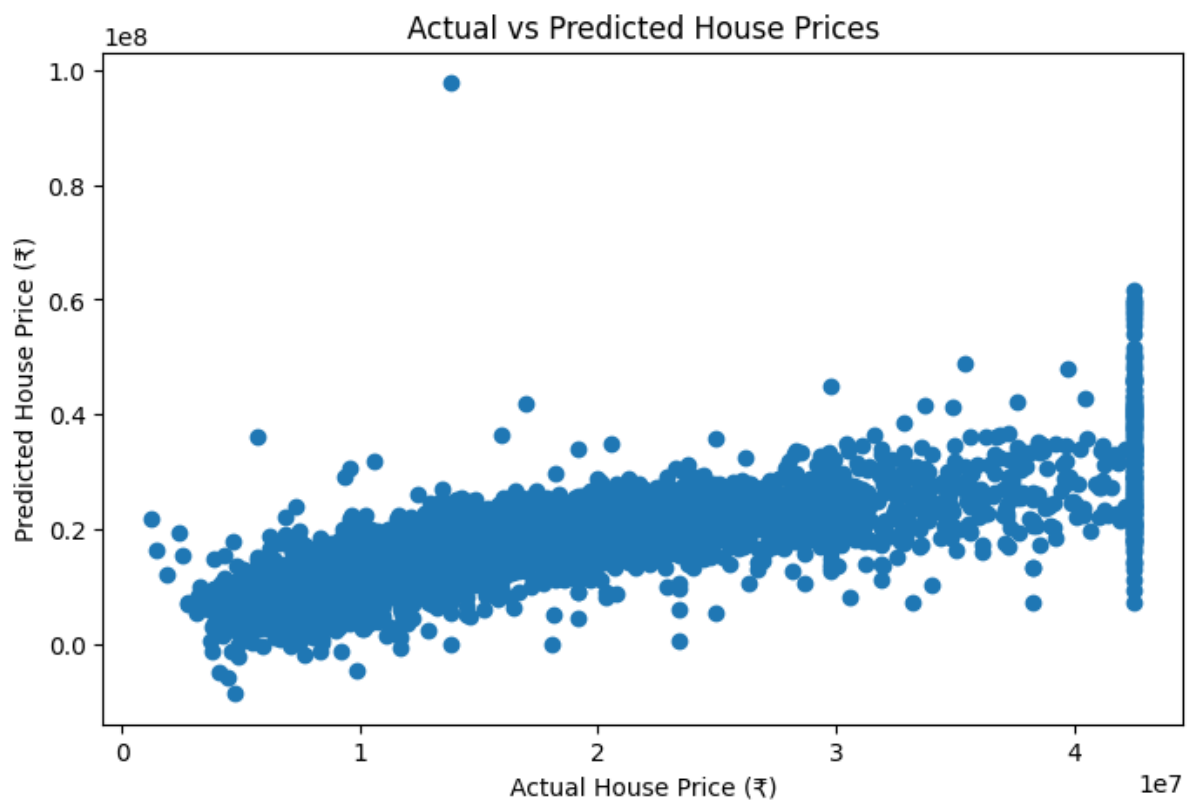
ACTUAL VS PREDICTED PRICE

Actual Price (₹) Predicted Price (₹)

0 4054500.0 6112544.15

1 3893000.0 14994140.85

2	42500085.0	23032100.08
3	18581000.0	24130870.41
4	23630000.0	22139586.64
5	13489500.0	17099906.22
6	16847000.0	22486750.42
7	13387500.0	18434420.23
8	28900000.0	23296344.75
9	37961000.0	33282725.16



NEW HOUSE PRICE PREDICTION

Predicted Price in USD : \$ 247310.87

Predicted Price in INR : ₹ 21021424.21